# Comprehensive Examination
## (EC-3 Regular)

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| Course No. | : SS ZG526 |
| Course Title | : DISTRIBUTED COMPUTING |
| Nature of Exam | : Open Book |
| Weightage | : 45% |
| Duration | : 3 Hours |
| Date of Exam | : 08/04/2017 (AN) |

No. of Pages = 3
*No. of Questions = 3*

Note:
1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1 (a) Consider the space-time diagram consisting of 4 processes shown in Fig. 1. Initially, the vector clocks of all the 4 processes are [0 0 0 0]. Using Singhal-Kshemkalyani's differential technique, determine the progress of vector clocks at each of the processes and the contents of the messages exchanged among them. [6]

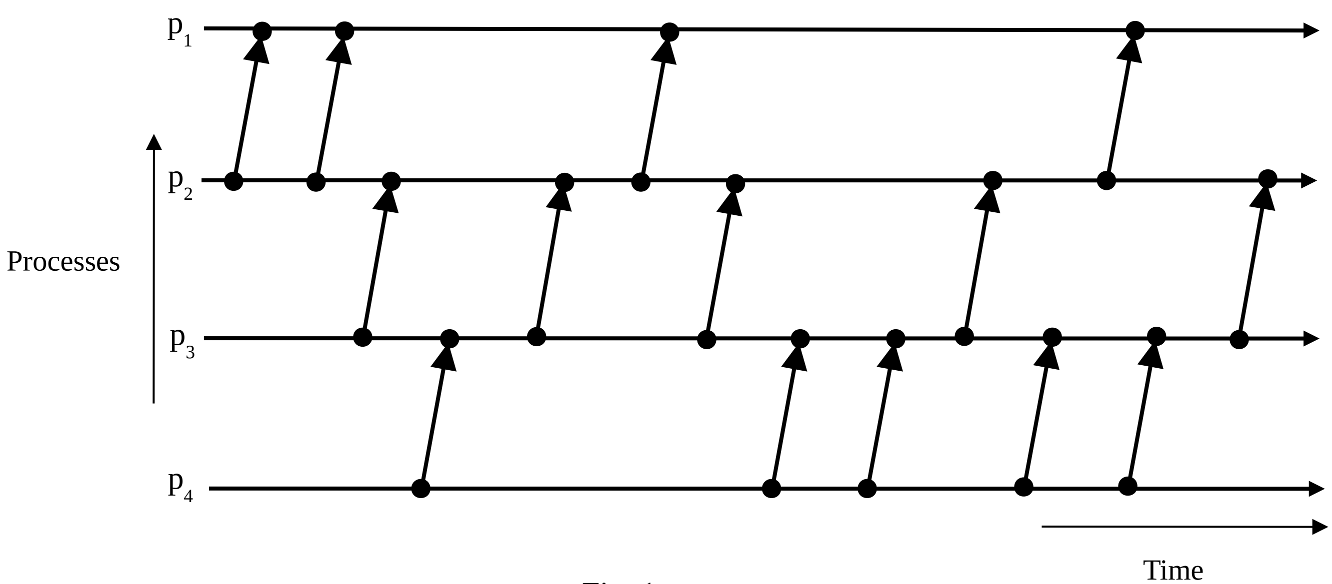


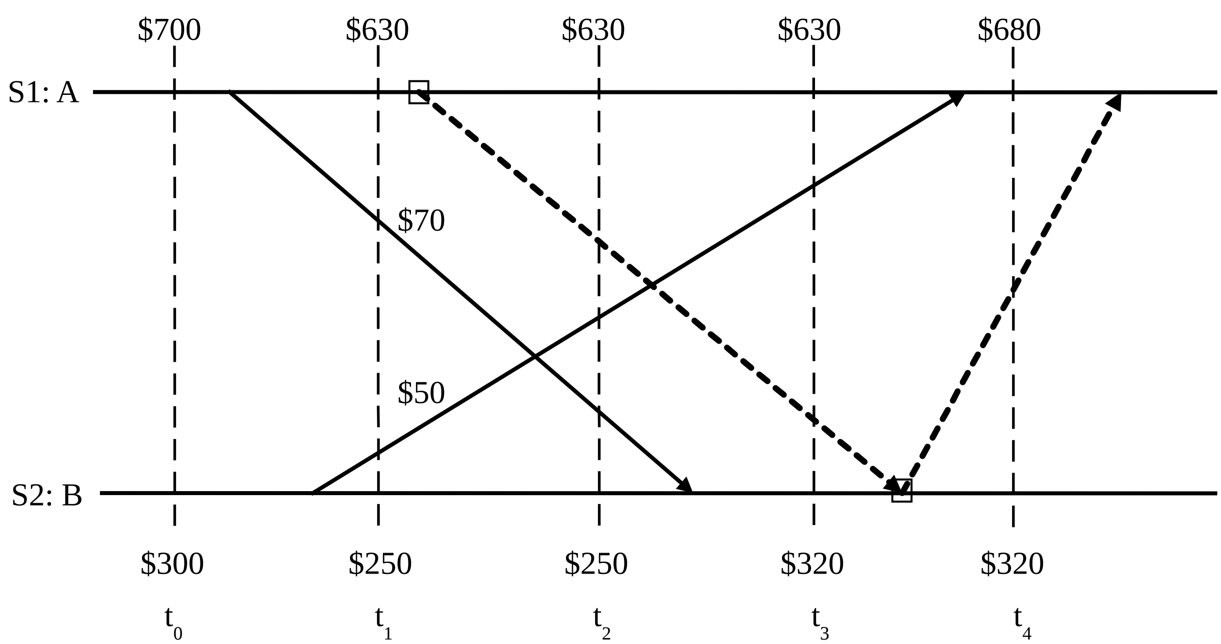

Fig. 1

Q.1 (b) Apply Fowler-Zwaenepoel's direct-dependency technique to determine the progress of vector clocks at each of the 4 processes shown in Fig. 1 and also the contents of the messages exchanged among them. Assume that initially, the vector clocks of all the 4 processes are [0 0 0 0]. [6]

Q.1 (c) Identify the events for which the vector clock values obtained by applying the two techniques differ? What is the reason behind this difference? [4 + 1 = 5]

Q.2 (a) Consider the money transfer scenario shown in the timing diagram of Fig. 2. S1 and S2 are two sites of a distributed system which maintain bank accounts A and B respectively. Communication channel from site S1 to site S2 is denoted as $C_{12}$ and the communication channel from site S2 to site S1 is denoted as $C_{21}$. The dashed vertical lines denote the different time instants of this timing diagram. Initially, at time instant $t_0$, account A has $700 and account B has $300. Between time instants $t_0$ and $t_1$, S1 initiates a transfer of $70 from account A to account B and S2 initiates a transfer of $50 from account B to account A (as shown by the 2 bold arrows in the diagram). S2 receives the money transfer message after $t_2$ and S1 receives the money transfer message after $t_3$. The states of channels $C_{12}$ and $C_{21}$ at the different time instants are also shown in the Fig. Assume that the Chandy-Lamport algorithm is executed on this distributed system for recording the global snapshot. S1 initiates the snapshot recording algorithm by sending a marker to S2 after time instant $t_1$ which S2 receives after time instant $t_3$ and sends a marker to S1 after $t_3$. S1 receives this marker after $t_4$. The sending and the receiving of markers are shown using dotted arrows in the Fig. Describe **how and when** the Chandy-Lamport algorithm records the local states of S1 and S2 and the states of the channels $C_{12}$ and $C_{21}$. Note that the amount of $1000 in the system should be conserved in the recorded global state.

[5]

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|---|---|---|---|---|---|
| $C_{12}$: | $0 | $70 | $70 | $0 | $0 |
| $C_{21}$: | $0 | $50 | $50 | $50 | $0 |

Fig. 2

Q.2 (b) Consider the following history of message exchange among the 4 processes $p_1$, $p_2$, $p_3$ and $p_4$.

- $p_1$ sent 3 messages to $p_2$
- $p_1$ sent 4 messages to $p_3$
- $p_1$ sent 2 messages to $p_4$
- $p_2$ sent 5 messages to $p_1$

- $p_2$ sent 2 messages to $p_3$
- $p_2$ sent 3 messages to $p_4$
- $p_3$ sent 4 messages to $p_1$
- $p_3$ sent 1 message to $p_2$
- $p_3$ sent 3 messages to $p_4$
- $p_4$ sent 2 messages to $p_1$
- $p_4$ sent 6 messages to $p_2$
- $p_4$ sent 5 messages to $p_3$

The following are the contents of the DELIV arrays for $p_1$, $p_2$, $p_3$ and $p_4$:

- for $p_1$, $DELIV_1$ = [0 5 3 1]
- for $p_2$, $DELIV_2$ = [2 0 1 4]
- for $p_3$, $DELIV_3$ = [3 2 0 4]
- for $p_4$, $DELIV_4$ = [2 2 1 0]

Now, $p_1$ sends message $m_1$ to $p_2$, $p_2$ sends message $m_2$ to $p_3$, $p_3$ sends message $m_3$ to $p_4$ and $p_4$ sends message $m_4$ to $p_1$. Using Raynal-Schiper-Toueg algorithm, determine for each of $m_1$, $m_2$, $m_3$ and $m_4$, whether buffering is required or not for maintaining causal ordering of messages and why. [4]

Q.3 (a) Suppose there are 11 nodes – N2, N8, N13, N23, N30, N58, N69, N75, N102, N115 and N123. For a Chord ring for m = 7, show with the help of a diagram, the arrangement of these nodes along the ring. Also, determine the node on which each of the following keys will be stored – K2, K10, K25, K60, K65, K71, K74, K80, K95 and K110. You should clearly mention the reason for placing each of the keys on the particular nodes.
[1 + 5 (½ x 10) = 6]

Q.3 (b) Suppose for the Chord ring obtained in Q.3 (a), a lookup for K110 is initiated at N58. Describe the step-by-step procedure by which K110 can be located using the **simple lookup algorithm**. [4]

Q.3 (c) For the Chord ring obtained in Q.3 (a), determine the finger tables for N8, N58 and N102.
[2 + 2 + 2 = 6]

Q.3 (d) For the same Chord ring of Q.3 (a), assume K110 is to be located using the **scalable lookup algorithm** and the search is initiated at N58. Describe the steps followed by the **scalable lookup algorithm** to locate K110. [2]

Q.3 (e) Why do we use a left-open right-closed interval in simple lookup and scalable lookup algorithms for searching keys in any Chord ring? [1]

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